

VAIBHAV VISWANATH

AI Systems & GPU Optimization Researcher

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DETAILS

Open to AI Systems / GPU Engineering Roles
Linux-Based Performance Optimization

SKILLS

Programming

C++, Python, C, Swift, Go, R

GPU & Acceleration

CUDA, TensorRT, cuDNN, DeepStream

Deep Learning

PyTorch, CNNs, RNNs, Transformers, GANs

Systems

Linux, NUMA Optimization, CPU Affinity
Memory Hierarchy, OS Scheduling

Core CS

Data Structures & Algorithms (500+ problems)
Computer Architecture
Operating Systems
Probability & Statistics

EDUCATION

B.Tech CSE (AI & ML)

VIT Vellore

GPA: 6.24

PROFILE

AI systems-focused engineer with research experience in OS-level inference optimization and GPU acceleration. Experienced in CUDA kernel development, TensorRT deployment, and performance benchmarking across CPU-GPU architectures. Strong foundation in computer architecture and distributed learning.

RESEARCH

Operating System-Level Optimization for Accelerating AI Inference

- Conducted NUMA-based CPU affinity experiments using `numactl`
- Benchmarked inference latency under memory binding configurations
- Analyzed scheduling impact on throughput
- Identified system-level bottlenecks affecting AI performance

Artificial Intelligence and Privacy Concerns (Published)

- Investigated privacy vulnerabilities in large-scale AI systems
- Proposed mitigation strategies for secure AI deployment

Suicide Risk Prediction using Machine Learning (Under Review)

- Designed predictive ML models with imbalance handling
- Evaluated performance using ROC-AUC and F1 metrics

PROJECTS

CUDA Kernel Optimization Suite

- Implemented naive and tiled CUDA kernels
- Achieved 240× speedup over CPU baseline

TensorRT Inference Acceleration

- Converted deep learning models to optimized TensorRT engines
- Reduced inference latency using FP16 precision

Multilingual Sign Language Detection

- YOLOv8 + MobileNetV2 pipeline (841 classes)
- GPU-optimized inference on NVIDIA T4